99.500 Applied Mathematics for Engineering

Homework 3 Solution

March 24, 2026

1. (a) Given Fourier's Law $q = -k\frac{dT}{dr}$, by shell balance equation in a sphere,

$$\left(4\pi r^2 q\right)\Big|_r - \left(4\pi r^2 q\right)\Big|_{r+\Delta r} = c \cdot \rho \cdot 4\pi r^2 \Delta r \frac{\partial T}{\partial t} \tag{1}$$

Divide both sides by Δr and take the limit when $\Delta r \to 0$ to obtain

$$\frac{\partial T}{\partial t} = \frac{k}{c\rho}\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial T}{\partial r}\right) \tag{2}$$

Let $\alpha = \frac{k}{c\rho}$ which is defined as thermal diffusivity to have

$$\frac{\partial T}{\partial t} = \frac{\alpha}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial T}{\partial r}\right) \tag{3}$$
$$\text{IC}: T(r,0) = T_0 \tag{4}$$
$$\text{BC1}: T(R,t) = T_\infty \tag{5}$$
$$\text{BC2}: \frac{\partial T}{\partial r}\Big|_{r=0} = 0 \tag{6}$$

Non-dimensionalization: $\theta = \frac{T-T_\infty}{T_0-T_\infty}$, $\eta = \frac{r}{R}$ and $\tau = \frac{\alpha t}{R^2}$ to transform the Equation (2) with IC and BCs into

$$\frac{\partial \theta}{\partial \tau} = \frac{1}{\eta^2}\frac{\partial}{\partial \eta}\left(\eta^2\frac{\partial \theta}{\partial \eta}\right) \tag{7}$$
$$\theta(\eta,0) = 1 \tag{8}$$
$$\theta(1,\tau) = 0 \tag{9}$$
$$\frac{\partial \theta}{\partial \eta}\Big|_{\eta=0} = 0 \tag{10}$$

Assume $\theta(\eta,\tau) = X(\eta)Y(\tau)$ and substitute into Equation (7)

$$\frac{1}{Y}\frac{dY}{d\tau} = \frac{\frac{1}{\eta^2}\frac{d}{d\eta}\left(\eta^2\frac{dX}{d\eta}\right)}{X} = -\lambda^2 \tag{11}$$

$$\frac{1}{Y}\frac{dY}{d\tau} = -\lambda^2 \implies Y = c_1 e^{-\lambda^2\tau} \tag{12}$$
$$\frac{\frac{1}{\eta^2}\frac{d}{d\eta}\left(\eta^2\frac{dX}{d\eta}\right)}{X} = -\lambda^2 \implies \eta^2 X'' + 2\eta X' + \lambda^2\eta^2 X = 0 \tag{13}$$
$$\implies X(\eta) = \eta^{-\frac{1}{2}}\left(c_2 J_{\frac{1}{2}}(\lambda\eta) + c_3 J_{-\frac{1}{2}}(\lambda\eta)\right) \tag{14}$$
$$\implies X(\eta) = c_4\frac{\sin(\lambda\eta)}{\eta} + c_5\frac{\cos(\lambda\eta)}{\eta} \tag{15}$$

as it is a generalized Bessel equation with $p = \frac{1}{2}$,$a = 2$, $b = 0$, $c = 0$, $s = 1$, $d = \lambda^2$.

$$\frac{\partial \theta}{\partial \eta}\Big|_{\eta=0} = 0 \implies X'(0) = 0 \tag{16}$$

Differentiate Equation (15), take $\eta = 0$ and from Equation (16) to conclude $c_5 = 0$. Hence,

$$X(\eta) = c_4 \frac{\sin(\lambda\eta)}{\eta} \tag{17}$$

Subsequently,

$$\theta(1,\tau) = 0 \Longrightarrow X(1) = 0 \Longrightarrow \sin(\lambda\eta) = 0 \Longrightarrow \lambda_n = n\pi,\ n = 1, 2, 3... \tag{18}$$

Combine Equations (12) and (17) to have

$$\theta(\eta,\tau) = \sum_{n=1}^{\infty} A_n \frac{\sin(n\pi\eta)}{\eta} e^{-(n\pi)^2\tau} \tag{19}$$

$$\theta(\eta,0) = 1 \Longrightarrow \sum_{n=1}^{\infty} A_n \frac{\sin(n\pi\eta)}{\eta} = 1 = f(x) \tag{20}$$

Equation (13) is S-L system with w$(\eta) = \eta^2$,

$$A_n = \frac{\int_a^b f(x)\phi_n(x)w(x)dx}{\int_a^b \phi_n^2(x)w(x)dx} = \frac{\int_0^1 \frac{\sin(n\pi\eta)}{\eta} \cdot \eta^2 d\eta}{\int_0^1 \left(\frac{\sin(n\pi\eta)}{\eta}\right)^2 \cdot \eta^2 d\eta} = \frac{2}{n\pi}(-1)^{n+1} \tag{21}$$

$$\Longrightarrow \theta(\eta,\tau) = \sum_{n=1}^{\infty} \frac{2}{n\pi}(-1)^{n+1} \frac{\sin(n\pi\eta)}{\eta} e^{-(n\pi)^2\tau} \tag{22}$$

$$\Longrightarrow \frac{T - T_\infty}{T_0 - T_\infty} = \sum_{n=1}^{\infty} \frac{2}{n\pi}(-1)^{n+1} \frac{\sin(n\pi\frac{r}{R})}{\frac{r}{R}} e^{-(n\pi)^2\frac{\alpha t}{R^2}} \tag{23}$$

(b) The average dimensionless temperature can be evaluated as

$$\theta_{average} = \frac{\int_0^1 4\pi\eta^2\theta d\eta}{\frac{4}{3}\pi 1^3} \tag{24}$$

$$= \frac{4\pi\int_0^1 \sum_{n=1}^{\infty} \frac{2}{n\pi}(-1)^{n+1}\eta\sin(n\pi\eta)e^{-(n\pi)^2\tau}d\eta}{\frac{4}{3}\pi} \tag{25}$$

$$= \sum_{n=1}^{\infty} \frac{6}{(n\pi)^2} e^{-(n\pi)^2\tau} \tag{26}$$

The sphere's average original temperature as a function of time is

$$\frac{T_{average} - T_\infty}{T_0 - T_\infty} = \frac{6}{\pi^2}\sum_{n=1}^{\infty} \frac{e^{-\frac{\alpha n^2\pi^2 t}{R^2}}}{n^2} \tag{27}$$

2. Assume $C(x,t) = R(x)T(t)$ and substitute into

$$\frac{\partial C}{\partial t} = \frac{1}{x}\frac{\partial}{\partial x}\left(x\frac{\partial C}{\partial x}\right) \tag{28}$$

to obtain

$$\frac{1}{T}\frac{dT}{dt} = \frac{1}{Rx}\frac{d}{dx}\left(x\frac{dR}{dx}\right) = -\lambda^2 \tag{29}$$

$$\frac{1}{T}\frac{dT}{dt} = -\lambda^2 \implies T(t) = Ae^{-\lambda^2 t} \tag{30}$$

$$\frac{1}{Rx}\frac{d}{dx}\left(x\frac{dR}{dx}\right) = -\lambda^2 \implies x^2R'' + xR' + \lambda^2x^2R = 0 \tag{31}$$

$$\implies R(x) = B_1J_0(\lambda x) + C_1Y_0(\lambda x) \tag{32}$$

$$\left.\frac{\partial C}{\partial x}\right|_{x=1} = 0 \implies R'(1) = 0 \tag{33}$$

$$\alpha\left.\frac{\partial C}{\partial x}\right|_{x=\beta} = C(\beta,t) \implies \alpha R'(\beta) = R(\beta) \tag{34}$$

From Equation (32) to have

$$R'(x) = -B_1\lambda J_1(\lambda x) - C_1\lambda Y_1(\lambda x) \tag{35}$$

$$R'(1) = -B_1\lambda J_1(\lambda) - C_1\lambda Y_1(\lambda) = 0 \implies C_1 = -\frac{J_1(\lambda)}{Y_1(\lambda)}B_1 \tag{36}$$

From Equation (34) to have

$$-\alpha B_1\lambda J_1(\lambda\beta) - \alpha C_1\lambda Y_1(\lambda\beta) = B_1J_0(\lambda\beta) + C_1Y_0(\lambda\beta) \tag{37}$$

$$\frac{B_1}{Y_1(\lambda)}\left\{\alpha\lambda\left[J_1(\lambda)Y_1(\lambda\beta) - J_1(\lambda\beta)Y_1(\lambda)\right] + \left[J_1(\lambda)Y_0(\lambda\beta) - J_0(\lambda\beta)Y_1(\lambda)\right]\right\} = 0 \tag{38}$$

Hence, λ_n is the n-th root of the equation

$$\alpha x = \frac{J_0(x\beta)Y_1(x) - J_1(x)Y_0(x\beta)}{J_1(x)Y_1(x\beta) - J_1(x\beta)Y_1(x)} \tag{39}$$

and

$$R_n(x) = B_{1n}J_0(\lambda_n x) + C_{1n}Y_0(\lambda_n x) \tag{40}$$

$$= B_{1n}\left[J_0(\lambda_n x) - \frac{J_1(\lambda_n)}{Y_1(\lambda_n)}Y_0(\lambda_n x)\right] \tag{41}$$

$$C(x,t) = \sum_{n=1}^{\infty} R_n(x)T_n(t) = \sum_{n=1}^{\infty} D_n\left[J_0(\lambda_n x) - \frac{J_1(\lambda_n)}{Y_1(\lambda_n)}Y_0(\lambda_n x)\right]e^{-\lambda_n^2 t} \tag{42}$$

$$C(x,t) = \sum_{n=1}^{\infty} D_n\left[J_0(\lambda_n x) - \frac{J_1(\lambda_n)}{Y_1(\lambda_n)}Y_0(\lambda_n x)\right] = 1 = f(x) \tag{43}$$

Equation (31) is S-L system with $w(x) = x$,

$$D_n = \frac{\int_a^b f(x)\phi_n(x)w(x)dx}{\int_a^b \phi_n^2(x)w(x)dx} = \frac{\int_1^\beta 1\cdot\left[J_0(\lambda_n x) - \frac{J_1(\lambda_n)}{Y_1(\lambda_n)}Y_0(\lambda_n x)\right]xdx}{\int_1^\beta\left[J_0(\lambda_n x) - \frac{J_1(\lambda_n)}{Y_1(\lambda_n)}Y_0(\lambda_n x)\right]^2 xdx} \tag{44}$$

The final solution is

$$C(x,t) = \sum_{n=1}^{\infty} D_n\left[J_0(\lambda_n x) - \frac{J_1(\lambda_n)}{Y_1(\lambda_n)}Y_0(\lambda_n x)\right]e^{-\lambda_n^2 t} \tag{45}$$

and λ_n is the n-th root of the equation: $\alpha\text{x}=\frac{J_0(x\beta)Y_1(x)-J_1(x)Y_0(x\beta)}{J_1(x)Y_1(x\beta)-J_1(x\beta)Y_1(x)}$

3. Assume $C(x,t) = R(x)T(t)$ and substitute into

$$\frac{\partial C}{\partial t} = \frac{1}{x}\frac{\partial}{\partial x}\left(x\frac{\partial C}{\partial x}\right) - \delta C \tag{46}$$

to obtain

$$\frac{1}{T}\frac{dT}{dt} = \frac{1}{Rx}\frac{d}{dx}\left(x\frac{dR}{dx}\right) - \delta \tag{47}$$

$$\frac{1}{T}\frac{dT}{dt} + \delta = \frac{1}{Rx}\frac{d}{dx}\left(x\frac{dR}{dx}\right) = -\lambda^2 \tag{48}$$

$$\frac{1}{T}\frac{dT}{dt} + \delta = -\lambda^2 \implies T = Ae^{-(\delta+\lambda^2)t} \tag{49}$$

$$\frac{1}{Rx}\frac{d}{dx}\left(x\frac{dR}{dx}\right) = -\lambda^2 \implies x^2R'' + xR' + \lambda^2x^2R = 0 \tag{50}$$

$$\implies R(x) = B_1J_0(\lambda x) + B_2Y_0(\lambda x) \tag{51}$$

$$\left.\frac{\partial C}{\partial x}\right|_{x=0} = 0 \implies R(0) = 0 \implies B_1J_0(0) + B_2Y_0(0) \implies B_2 = 0 \tag{52}$$

$$T(1,t) = 0 \implies R(1) = 0 \implies B_1J_0(\lambda) = 0 \implies J_0(\lambda_n) = 0 \tag{53}$$

where λ_n is the n-th root of $J_0(x) = 0$. Hence,

$$C(x,t) = \sum_{n=1}^{\infty} R_n(x)T_n(t) = \sum_{n=1}^{\infty} C_nJ_0(\lambda_n x)e^{-(\delta+\lambda_n^2)t} \tag{54}$$

$$C(x,0) = 1 \implies \sum_{n=1}^{\infty} C_nJ_0(\lambda_n x) = 1 = f(x) \tag{55}$$

Equation (50) is S-L system with $w(x) = x$,

$$C_n = \frac{\int_a^b f(x)\phi_n(x)w(x)dx}{\int_a^b \phi_n^2(x)w(x)dx} = \frac{\int_0^1 1\cdot J_0(\lambda_n x)\cdot x\cdot dx}{\int_0^1 J_0^2(\lambda_n x)\cdot x\cdot dx} = \frac{2}{\lambda_n J_1(\lambda_n)} \tag{56}$$

since

$$\int_0^1 1\cdot J_0(\lambda_n x)\cdot x\cdot dx = \frac{J_1(\lambda_n)}{\lambda_n} \tag{57}$$

$$\int_0^1 J_0^2(\lambda_n x)\cdot x\cdot dx = \frac{J_1^2(\lambda_n)}{2} \tag{58}$$

The final solution is

$$C(x,t) = \sum_{n=1}^{\infty} \frac{2}{\lambda_n J_1(\lambda_n)} J_0(\lambda_n x)e^{-(\delta+\lambda_n^2)t} \tag{59}$$